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Accelera



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Abstract

This report explains my individual work in three parts of the Accelera project. The first part is Auto Preprocessing, which prepares tabular, text, and image datasets before model training. The second part is the Accelera Pipeline reporting part, which generates reports for pipeline graphs, branches, and metrics. The third part is the Benchmarking module, which allows prediction submissions to be compared using the same dataset and evaluation metric. In this report, I describe the main idea, the steps I followed, the generated reports, the benchmark workflow, and the testing results.

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List of Abbreviations

AI	Artificial Intelligence
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbors
ML	Machine Learning
NLP	Natural Language Processing
PDF	Portable Document Format
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

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Chapter 1: Introduction

1.1 Motivation and Justification

In many machine learning projects, building the model is not the only important step. Before training the model, the dataset usually needs many preparation steps such as cleaning, checking missing values, splitting the data, encoding categorical columns, scaling numerical values, and making sure there are no errors. These steps are repeated in most projects, so doing them manually every time can take a lot of time and may lead to mistakes.

Accelerera tries to make these steps easier by providing them in one framework. The Auto Preprocessing module is responsible for preparing the data automatically. It applies the needed preprocessing steps and saves the fitted objects, such as encoders and scalers, so the same steps can be used later on new data. This is important because using different preprocessing steps between training and testing can give wrong results.

The Accelerera Pipe module helps the user evaluate and understand pipeline results in a clearer way. Metric nodes calculate the evaluation results during the pipeline flow, while the reporting part shows the generated graphs, explains the different branches, and allows the user to compare the results of these branches. Instead of only displaying numbers in the terminal, the module creates HTML reports that include graphs, tables, and metric summaries.

The Benchmarking module is used when different users want to compare their results on the same task. The benchmark creator prepares the test data and keeps the correct target values hidden. Then, users submit their predictions, and the system evaluates them using the same metric. This makes the comparison fair because all submissions are tested using the same rules.

1.2 Project Objectives and Problem Definition

In machine learning projects, preparing the data and understanding the results are important steps before and after model training. However, these steps may become difficult when they are done manually, especially when the dataset contains different types of data or when the user needs to repeat the same process many times.

This project focuses on improving these steps inside Accelerera. It aims to support the user during data preprocessing, pipeline execution, pipeline reporting, and benchmarking. The Auto Preprocessing module prepares different types of datasets for machine learning tasks. The Accelerera Pipe module includes metric nodes for calculating evaluation results and reporting classes for presenting pipeline outputs in a clear form using reports, graphs, tables, and metric summaries. The Benchmarking module gives users a fair way to submit their predictions and compare their results using the same evaluation rules.

the main objective of this work is to make the machine learning workflow more organized, easier to understand, and less dependent on manual steps.

1.3 Project Outcomes

The implemented work produces three main outcomes. First, the Auto Preprocessing module prepares tabular, text, and image datasets and saves the fitted artifacts needed for later testing. Second, the Accelerera Pipe module calculates metric results through metric nodes and generates readable pipeline outputs using HTML pages, tables, graphs, and metric summaries. Third, the Benchmarking module supports benchmark creation, submission validation, scoring, and leaderboard display.

1.4 Document Organization

Chapter 2 summarizes the related concepts and previous work. Chapter 3 explains the system design and architecture of Auto Preprocessing, the Accelerera Pipe module, and Benchmarking. Chapter 4 presents testing and comparative evaluation. Chapter 5 concludes the report and lists future improvements. The user guide is in the appendix.

Chapter 2: Literature Survey

In this chapter, I give an overview of the topics, algorithms, and techniques needed to understand my work in the project. I discuss the technical subjects and background knowledge.

2.1 Background and Previous Work

This section introduces the main background needed to understand these parts in the project. It explains the important concepts, techniques, and tools that were used in different parts of the system. These background topics help clarify how the project supports data preparation, pipeline execution, report generation, model evaluation, and benchmarking. The goal of this section is to give the reader enough context before moving to the methodology and implementation chapters.

2.1.1 Data Science and Machine Learning

Data science is the field that uses data, statistics, programming, and domain knowledge to understand a problem and extract useful insights from it. It is usually used when we have raw data and we want to turn it into useful information or predictions.

Machine learning is one of the main parts of data science. Instead of writing fixed rules for every case, we train a model on examples, and the model learns patterns from these examples. After training, the model can use what it learned to make predictions on new data.

In supervised machine learning, the training data contains input features and a target label. The model learns the relationship between the features and the target. If the target is a class, such as spam or not spam, the problem is called classification. If the target is a continuous number, such as price or temperature, the problem is called regression.

2.1.1.1 Data Exploration and Preprocessing [1]

Exploratory Data Analysis, or EDA, is the step where we inspect the dataset before building the model. We summarize columns, draw visualizations, check missing values, and try to understand relationships between variables. This step helps us notice problems early instead of discovering them after training the model.

Data preprocessing is the stage of transforming raw data into a clean and usable format for machine learning models. In real datasets, the data may contain missing values, duplicated rows, different feature scales, and categorical columns. Because of this, preprocessing is usually considered an essential part of a machine learning workflow.

Missing value imputation is used when some values in a dataset are empty. Instead of removing all rows that contain missing values, the missing values can be replaced using a suitable value, such as the mean, median, mode, or another estimated value. Robust scaling is a scaling technique that uses the median and the interquartile range instead of the mean and standard deviation. It is useful when the data contains outliers, because the median and interquartile range are less affected by extreme values:

$$x' = \frac{x - \text{median}(X)}{\text{IQR}(X)}$$

where x' is the scaled value, x is the original value, $\text{median}(X)$ is the median of the feature values, and $\text{IQR}(X)$ is the interquartile range.

Categorical encoding is the process of converting text categories into numeric values. Many traditional machine learning models and preprocessing pipelines expect numerical input. Two examples are binary encoding and frequency encoding. Binary encoding first gives each category a number, then converts this number into binary form and stores the binary digits in separate columns. Frequency encoding replaces each category with how often it appears in the dataset:

$$f(c) = \frac{\text{count}(c)}{N}$$

where $f(c)$ is the frequency of category c , $\text{count}(c)$ is the number of rows containing this category, and N is the total number of rows.

2.1.1.2 Model Evaluation Metrics [2]

After training a model, we need metrics to check whether the model is performing well or not. The selected metric depends on the type of problem. Classification models are evaluated using metrics that compare predicted classes with actual classes, while regression models are evaluated using metrics that measure the difference between predicted numerical values and real values.

Classification metrics: Classification metrics are used when the output is a class or category, such as spam or not spam, positive or negative, or one image class from many classes. The confusion matrix is the base idea behind many classification metrics. It compares the actual classes with the predicted classes. For binary classification, it contains four values: true positives, true negatives, false positives, and false negatives.

- **True Positive (TP):** The model predicts positive and the actual class is positive.
- **True Negative (TN):** The model predicts negative and the actual class is negative.
- **False Positive (FP):** The model predicts positive but the actual class is negative.
- **False Negative (FN):** The model predicts negative but the actual class is positive.

Accuracy measures how many predictions are correct out of all predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives. Accuracy is simple and useful when the classes are balanced, but it may be misleading when one class is much larger than the other.

Precision measures how many predicted positive samples are actually positive:

$$Precision = \frac{TP}{TP + FP}$$

Precision is useful when false positives should be reduced.

Recall measures how many actual positive samples were correctly detected:

$$Recall = \frac{TP}{TP + FN}$$

Recall is useful when false negatives should be reduced.

The F1-score combines precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

It is useful when both precision and recall are important, especially with imbalanced datasets.

Regression metrics: Regression metrics are used when the output is a numerical value, such as price, score, or temperature.

Mean Absolute Error measures the average absolute difference between the real values and predicted values:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of samples. MAE is easy to understand because it gives the average error in the same unit as the target value.

Mean Squared Error gives more penalty to large errors:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This means large mistakes affect the score more than small mistakes.

Root Mean Squared Error is the square root of MSE:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE is easier to read than MSE because it returns the error to the same unit as the target value.

R-squared score measures how well the regression model explains the variation in the actual values:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the actual values. A value close to 1 means the model explains the data well, while a value close to 0 means the model does not explain the data well.

2.1.2 Natural Language Processing (NLP) [3]

Natural Language Processing is the field that helps computers understand and work with human language. It is used in many applications such as text classification, sentiment analysis, spam detection, and chatbots. Text data cannot be used directly by most machine learning models, so it needs to be cleaned and converted into numbers first.

2.1.2.1 Term Frequency-Inverse Document Frequency (TF-IDF)

Term Frequency-Inverse Document Frequency is one common technique used to convert text into numerical features. It gives each word a weight based on how important it is in one document and across the full dataset. Term Frequency measures how often a word appears in a document. Inverse Document Frequency gives lower weight to words that appear in many documents, because these words are usually less useful for classification:

$$TFIDF(t, d) = TF(t, d) \times IDF(t)$$

where $TFIDF(t, d)$ is the weight of term t in document d , $TF(t, d)$ is the frequency of the term in the document, and $IDF(t)$ measures how rare the term is across all documents.

2.1.3 Data Augmentation

Data augmentation is a technique used to increase the diversity of a dataset without actually collecting new data. It works by applying transformations to existing samples to create new modified versions. For example, in image classification, augmentation may include flipping, rotating, cropping, or changing brightness. This can help the model

generalize better because it sees more variations during training.

2.2 Comparative Study of Previous Work

The most direct external comparison in this report is AutoCleanML because it also automates common tabular preprocessing tasks. The comparison focuses on the effect of preprocessing on model results, using the same validation split and the same machine learning models where possible.

2.3 Implemented Approach

For the preprocessing part, I used a classical automated preprocessing approach. I chose this approach because it is simple, fast, and easy to understand compared to more complex model based preprocessing methods. It also fits the goal of the module, which is to prepare the data before model training without adding heavy computation. The tabular workflow depends on common preprocessing techniques such as missing value imputation, `RobustScaler`, categorical encoding, frequency encoding, and mutual information feature selection.

I also extended the preprocessing idea so it would not work with tabular data only. For text data, I used cleaning, tokenization, stemming, and TF IDF feature extraction. For image data, I used folder based class detection, image validation, resizing, optional augmentation, and PyTorch `DataLoader` creation. Another important part of the approach is saving the fitted objects, such as transformers, selected features, dropped columns, target preprocessors, and class mappings. This makes the testing phase more consistent because new data can pass through the same steps used during training.

For pipeline reporting, I used an HTML report approach instead of depending only on terminal output. This was useful because pipeline results can include graphs, branches, selected paths, and different metric result types. Putting these outputs in one report makes the result easier to read and compare. The report was also modified to read the serialized pipeline graph, extract the branches, highlight selected nodes, and display metric results in a suitable format depending on whether the result is a number, array, dictionary, string, tuple, or figure.

For benchmarking, I used a hidden target evaluation approach. I chose this approach because it keeps the comparison fair between users. The benchmark keeps the correct target values hidden, and all submitted predictions are evaluated using the same test data, target file, and metric. I also added leaderboard direction handling, so the system can sort results correctly for metrics where higher is better and metrics where lower is better.

Chapter 3: System Design and Architecture

This chapter explains the design of the modules included in this individual report. The main modules are Auto Preprocessing, the Accelera Pipe module, and the Benchmarking Platform.

3.1 Overview and Assumptions

The design of these modules depends on some assumptions.

3.1.1 Auto Preprocessing Assumptions

For the Auto Preprocessing module, the assumptions depend on the type of dataset being processed. The module supports three types of data: tabular, text, and image datasets.

- **Data assumption:** The input data should be valid, meaningful, and one of the supported types, so the module can select the correct preprocessing workflow.
- **Configuration assumption:** The user should provide the needed configuration values, so the module can apply the correct preprocessing steps.

3.1.2 Benchmark Assumptions

For the Benchmark module, the following assumptions are made. These assumptions are introduced due to system design constraints, particularly the limited server storage capacity for handling large dataset uploads.

- **Benchmark data assumption:** The user is assumed to provide the benchmark files correctly, including the training data, test data, and separate target file. The user is also responsible for ensuring that the file separation does not cause data leakage, so the platform can evaluate submissions fairly.
- **Metric and submission assumption:** The admin is assumed to define the correct metric information for the benchmark, and the user is assumed to submit predictions in the required format, so the scoring script can calculate the result correctly.

3.1.3 Accelera Pipe Module Assumptions

The Accelera Pipe module depends on the following assumptions:

- **Metric node assumption:** The pipeline is assumed to pass the correct prediction output to the metric node. For supervised metrics, the true target values must be available. For unsupervised metrics, the input data used by the metric must be available.
- **Metric result assumption:** The metric results are assumed to be already generated from the pipeline modules and stored in a supported format, such as a number, array, dictionary, string, or tuple, so the report generator can read and display them correctly.
- **Visualization and output assumption:** If the result needs to be displayed as a figure, the user is assumed to provide a suitable plotting function that returns a Matplotlib `plt` object, and the output folder path should be valid so the generated report and images can be saved correctly.

Chapter 4: System Architecture

This section explains the implemented modules in this report. It discusses the three main parts: Auto Preprocessing, the Accelera Pipe module, and Benchmarking. For each module, the section explains module overview and how it works.

4.1 Auto Preprocessing Module

4.1.1 Module Overview

The Auto Preprocessing module automates the data understanding, exploratory data analysis, and preprocessing stages in the data science workflow. It helps reduce the repeated manual work needed to prepare datasets before modeling.

The module also helps the user understand the dataset before and after preprocessing by generating a report. This report shows important information about the input data and the preprocessing steps that were applied. For tabular data, the report shows information such as the number of rows and columns, data types, missing values, duplicate records, numerical and categorical features, basic statistics and EDA . For text data, it shows information such as the number of rows, target classes, class distribution, and number of missing. For image data, it shows information such as the number of images, image classes, class distribution, image sizes, and corrupted images. After preprocessing, the report shows the steps applied to the data, such as imputation, encoding, scaling, feature selection, text cleaning, TF IDF feature extraction, image resizing, normalization, and image preparation for classification.

Another important design point is separating the training phase from the inference phase. In the training phase, the preprocessing objects are fitted on the training data and saved as artifacts. In the inference phase, these saved artifacts are loaded and applied to new data. This ensures that new data is processed in the same way as the training data before it is passed to the trained model for prediction.

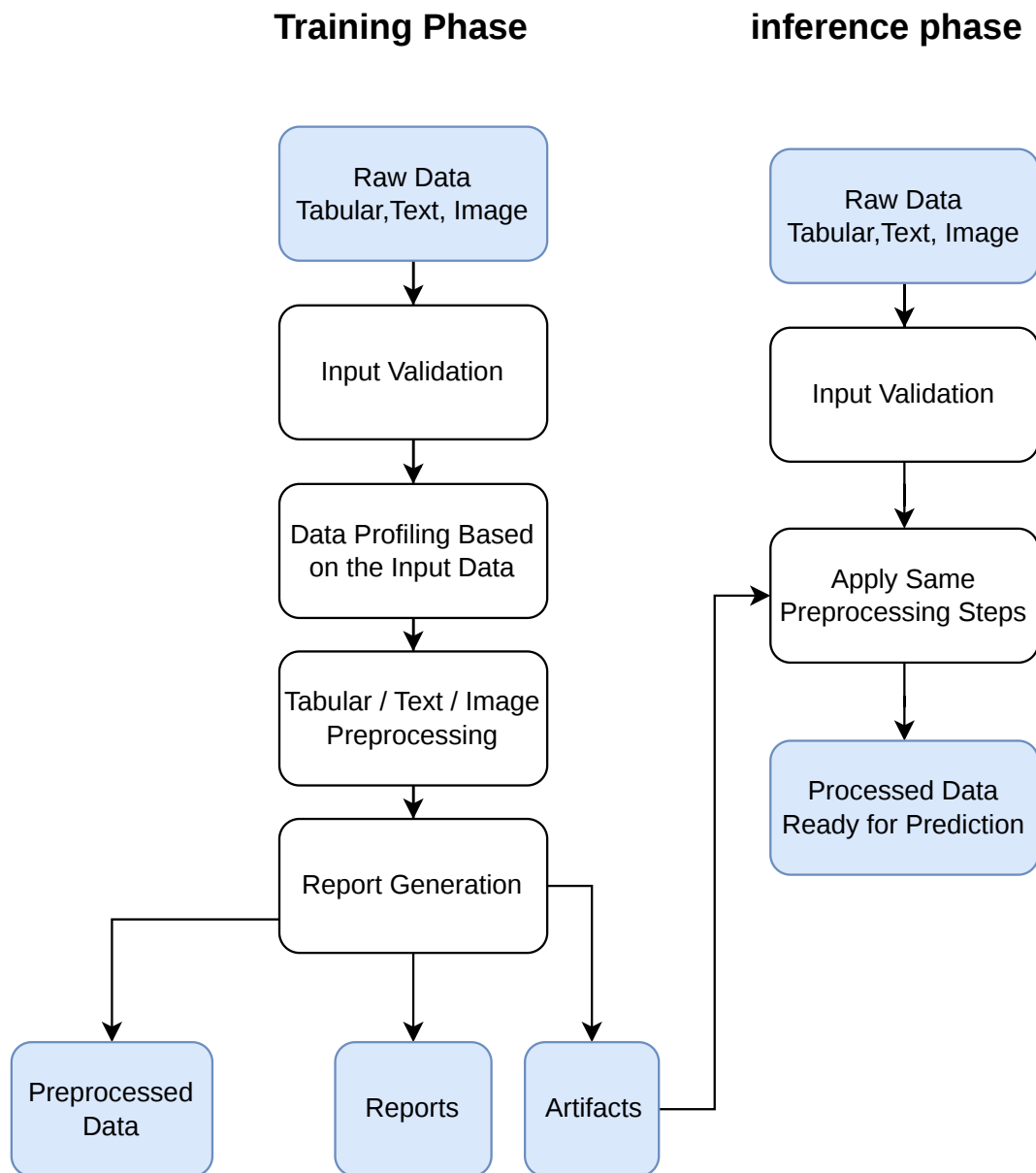


Figure 4.1: Auto Preprocessing Workflow

Supported Data Types and Task Handling

- tabular datasets for classification and regression
- text datasets for classification
- image datasets for classification

4.1.2 Tabular Training Flow

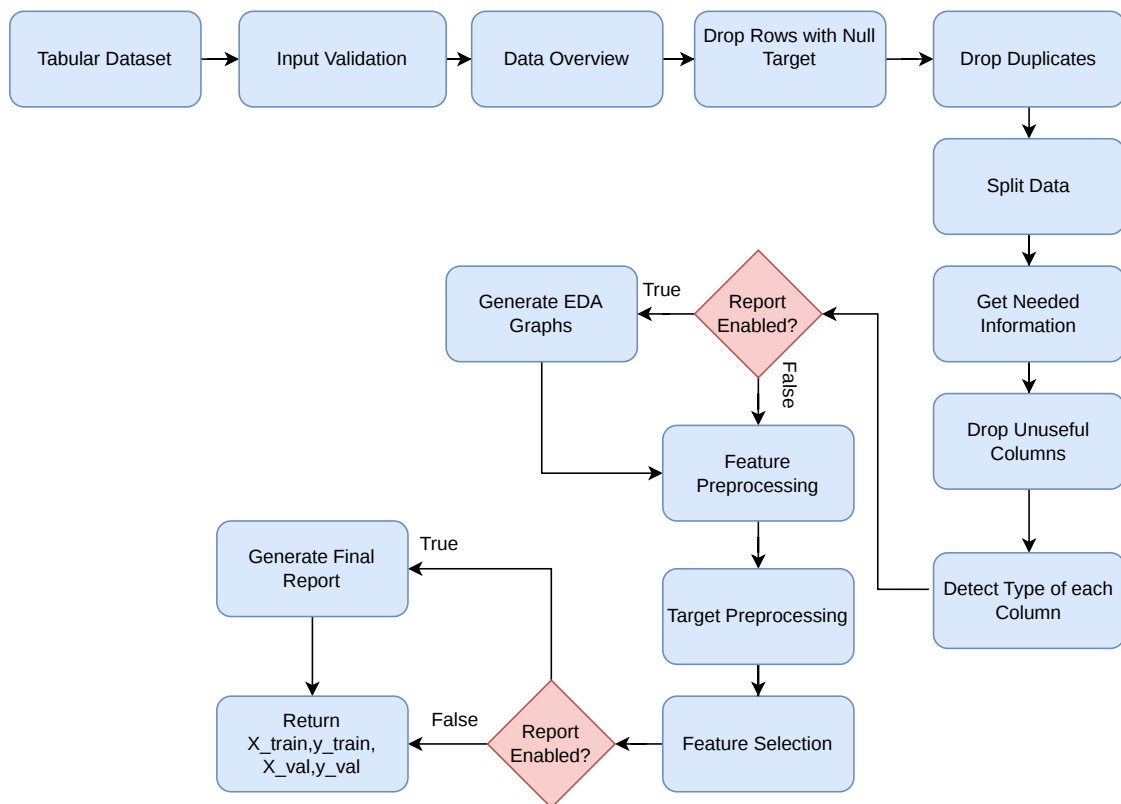


Figure 4.2: Tabular preprocessing workflow

The tabular training flow is implemented mainly in `classical_training_preprocessing.py`. The class receives a pandas DataFrame, target column, problem type, folder path, validation size, random state, thresholds, and report flag.

The training flow starts by validating the configuration. It checks the problem type, target column, validation size, random state, maximum ordinal threshold, cardinality threshold, missing value threshold, and feature importance threshold. It also saves the original input columns in `data_columns.pkl`, because the testing stage needs to compare new data against the same training structure.

The training flow is:

1. validate the input configuration
2. save original data columns for later testing
3. create a data overview for the report
4. remove rows has null target
5. remove duplicate rows
6. split data into training and validation sets

7. detect columns that should be dropped
8. detect each column type
9. generate graphs if reports are enabled
10. build a column transformer for feature preprocessing
11. preprocess the target according to classification or regression
12. apply feature selection using mutual information
13. save all fitted objects and metadata
14. generate the HTML report.

Algorithm 4.1: General tabular preprocessing flow

Input: DataFrame D, target column y, problem type, other configurations
 Output: X_train, X_val, y_train, y_val

- 1: Validate configuration
 - 2: Store original dataset columns
 - 3: Build data overview and report information
 - 4: remove rows has null target
 - 5: Remove duplicate rows
 - 6: Split into training and validation sets
 - 7: Drop user selected, high missing, constant
 - 8: Detect column types from training data
 - 9: Generate feature and target graphs when report is enabled
 - 10: Fit feature preprocessing objects on X_train
 - 11: Transform X_train and X_val
 - 12: Fit target preprocessing object on y_train
 - 13: Transform y_train and y_val
 - 14: Select useful features using mutual information
 - 15: Save fitted objects and metadata
 - 16: Generate report if enabled
-

4.1.2.1 Data Overview and Cleaning

Before applying feature transformations, the module creates a detailed data overview. The original first rows are stored, then all string values are converted to lowercase and another preview is stored. This helps categorical and text like values become more consistent before splitting and encoding.

The overview also records dataframe information, numerical statistics, categorical statistics, missing values per column, duplicate count, duplicate percentage, and dataset shape. After this, rows with missing target values are removed, because the model cannot learn correctly without a target label. Then duplicate rows are dropped and the new shape is stored in the report data.

For classification tasks, the split is stratified when the target has more than one class and contains no null values. The goal is to keep the class distribution closer between training and validation data. For other cases, the normal `train_test_split` function is used with the configured `val_size` and `random_state`.

4.1.2.2 Column Type Detection

The module assigns a type to each feature and then chooses preprocessing based on the type. This rule based design is easy to understand and works without writing special code for every dataset.

For every column, the module calculates the data type, number of unique values, unique value ratio, missing value count, missing value ratio, constant value status, mode, and for numerical columns the minimum, maximum, median, and mean. A feature can be dropped if the user listed it in `columns_need_to_drop`, if it is constant, or if its missing value ratio is greater than `missing_threshold`.

Table 4.1: Detected tabular column types and preprocessing

Column type	Detection rule	Applied preprocessing
Binary	columns with exactly two unique values.	Most frequent imputation followed by ordinal encoding with unknown values mapped to -1.
Ordinal	Integer columns with unique values less than or equal to <code>max_unique_ordinal</code> .	Most frequent imputation followed by robust scaling.
Numerical	Integer columns with unique values greater than <code>max_unique_ordinal</code> .	Median imputation followed by robust scaling.
Continuous	Floating point columns.	Median imputation followed by robust scaling.
Low cardinality categorical	Object columns with unique values less than or equal to <code>cardinality_threshold</code> .	Most frequent imputation followed by binary encoding.
High cardinality categorical	Object columns with unique values greater than <code>cardinality_threshold</code> .	Most frequent imputation followed by frequency encoding.
Unsupported or other	Any unsupported data type.	Removed from the preprocessing pipeline.

4.1.2.3 EDA Graphs

The EDA part is generated per column, not as one general graph for the whole dataset. In `make_graphs`, the module loops over the training feature columns after their types are detected. For each column, it chooses the correct graph wrapper based on both the column type and the problem type. After all feature graphs are generated, it adds a target graph and a numeric correlation matrix.

Table 4.2: EDA graphs generated for tabular preprocessing

Column type	Problem type	Generated graphs
Binary and categorical	Classification	Null vs non null pie chart, category count plot, and category count split by target class.
Binary and categorical	Regression	Null vs non null pie chart, category count plot, and target box plot for each category.
Ordinal	Classification	Null vs non null pie chart, ordered count plot, and ordered count plot split by target class.
Ordinal	Regression	Null vs non null pie chart, ordered count plot, and target box plot for each ordinal value.
Numerical and continuous	Classification	Null vs non null pie chart, histogram with KDE, and feature box plot grouped by class.
Numerical and continuous	Regression	Null vs non null pie chart, histogram with KDE, and scatter plot between feature and target.
Target column	Classification	Null vs non null pie chart and target class count plot.
Target column	Regression	Null vs non null pie chart, target histogram with KDE, and target box plot.
Numeric columns	Both	Correlation heatmap.

For categorical columns with many values, the graph wrappers keep the top five categories and group the remaining values as `Other`. This keeps the graphs readable and avoids a very crowded x axis.

4.1.2.4 Target Processing

Target null rows are removed before the data split. After feature preprocessing, the target is processed according to the problem type. For classification, the target labels are encoded using `LabelEncoder`. For regression, the target values are scaled using `RobustScaler`. The fitted target preprocessor is saved as `target_preprocessor.pkl`, and target metadata is saved in `target_info.pkl`. In testing, this metadata is used to apply the same target transformation when labels are provided.

4.1.2.5 Feature Selection

After feature preprocessing, the module applies mutual information based feature selection. For classification it uses mutual information for classification tasks, and for regression it uses mutual information for regression tasks. The selected features are saved in `selected_features.pkl`. This reduces unnecessary feature columns and keeps the testing stage aligned with training.

4.1.2.6 Saved Artifacts

Table 4.3: Saved artifacts for tabular preprocessing

Artifact	Purpose
<code>data_columns.pkl</code>	Original input columns before preprocessing.
<code>col_drop.pkl</code>	Columns removed during training preprocessing.
<code>training_preprocessor.pkl</code>	Fitted feature transformer.
<code>feature_names.pkl</code>	Names of generated transformed features.
<code>target_preprocessor.pkl</code>	Fitted label encoder or target scaler.
<code>target_info.pkl</code>	Target metadata used during testing.
<code>bool_type_col.pkl</code>	Boolean columns that need consistent handling.
<code>selected_features.pkl</code>	Feature names selected after mutual information.

4.1.3 Tabular Inference Flow

The testing class loads the saved artifacts and applies the same transformations to new data. The testing data must not fit a new preprocessor. It should only use what was fitted during training. The test data is checked against the saved column list, dropped columns are removed, feature preprocessing is applied, selected features are kept, and the target is transformed if it exists.

4.1.4 Text Training Flow

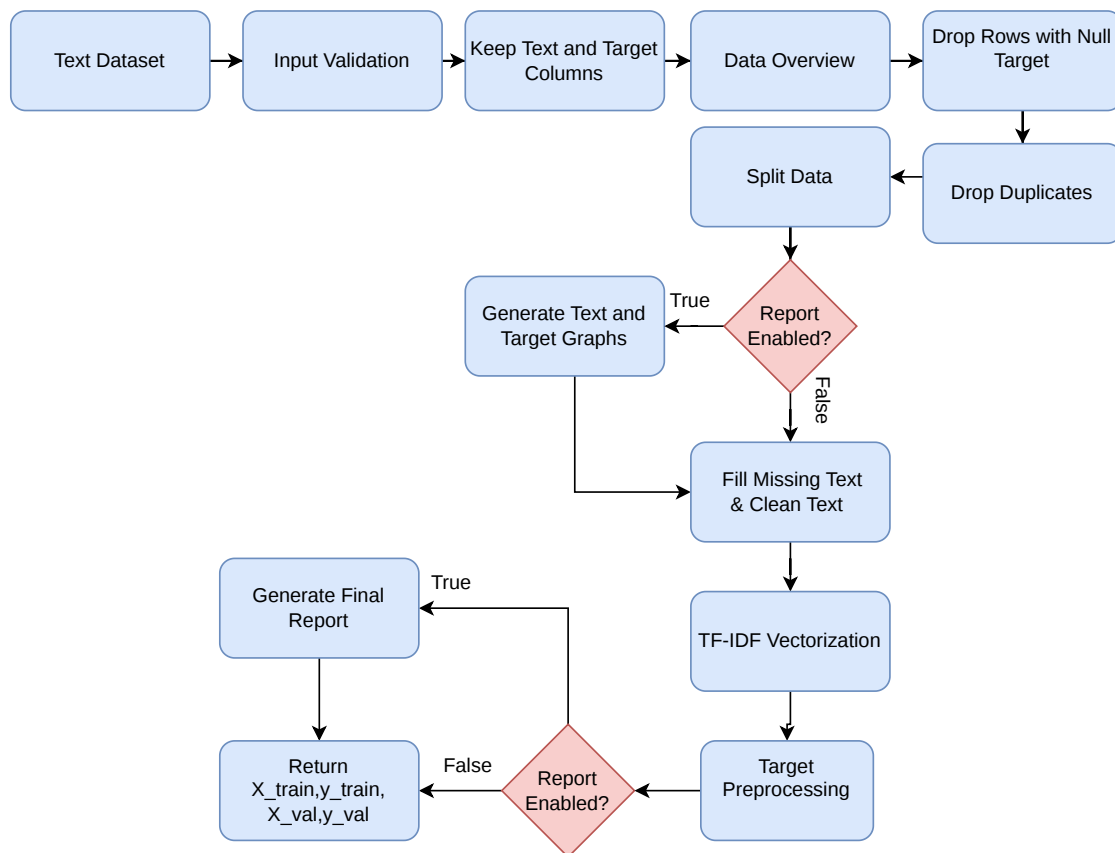


Figure 4.3: Text preprocessing workflow

The text preprocessing class is used when the dataset contains a text column and a target column. It inherits shared tabular behavior for overview, duplicates, and splitting, but it uses text specific feature preprocessing. During initialization, the class validates that the text column exists, that it is different from the target column, that the text column has object type, and that the target is suitable for classification. Any columns other than the text column and target column are removed because the text pipeline is designed for one text feature.

The report graphs for text are simpler than tabular column graphs. The `TextGraph` wrapper creates a null vs non null pie chart for the text column and a top word frequency plot. The target graph is generated using the classification target wrapper, so the user can also see the class distribution.

4.1.4.1 Text Tokenization

The custom tokenizer lowercases the text, handles common contractions, keeps important negation words such as `not` and `no`, removes noise characters, separates digits from letters, removes short tokens, and applies Porter stemming. This prepares the text before TF-IDF vectorization. The fitted `TfidfVectorizer` is saved as

`training_preprocessor.pkl`, while the target label encoder and metadata are saved for test time reuse.

The tokenizer also normalizes special apostrophe characters and expands words such as `can't`, `won't`, and `shan't`. This detail matters because negation can change the meaning of a sentence. For example, removing the word `not` can make a negative review look positive, so the stopwords list keeps `not` and `no`.

Algorithm 4.2: Text tokenizer flow from the implementation

Input: raw text sentence

Output: cleaned token list

- 1: Convert text to lowercase and remove extra spaces
 - 2: Normalize apostrophe characters
 - 3: Expand common negative contractions
 - 4: Add spaces between letters and digits
 - 5: Remove characters except letters, digits, and apostrophes
 - 6: Split text into words
 - 7: Remove stopwords except "not" and "no"
 - 8: Remove very short tokens
 - 9: Apply Porter stemming
 - 10: Return cleaned tokens
-

After cleaning, the module fits TF-IDF on the training text only and transforms the validation text with the same vectorizer. This avoids fitting on validation data. The report stores a small before and after preview, where the original text is shown beside the cleaned and stemmed tokens. For the target, the module uses `LabelEncoder` and saves it as `target_preprocessor.pkl`. It also saves `data_info.pkl` with the text column, and target column so the testing class can reuse the same configuration.

4.1.5 Text Inference Flow

The text testing class loads `data_info.pkl`, `training_preprocessor.pkl`, and `target_preprocessor.pkl`. It checks that the test dataframe contains the saved text column. If the target column is missing, it works in feature only mode and returns transformed features with `None` labels.

The testing flow lowercases the test data, fills missing text values with an empty string, and applies the saved TF-IDF vectorizer. If labels exist it transformed with the saved label encoder. This keeps the testing behavior close to training and prevents the test set from fitting its own vocabulary or label mapping.

4.1.6 Image Training Flow

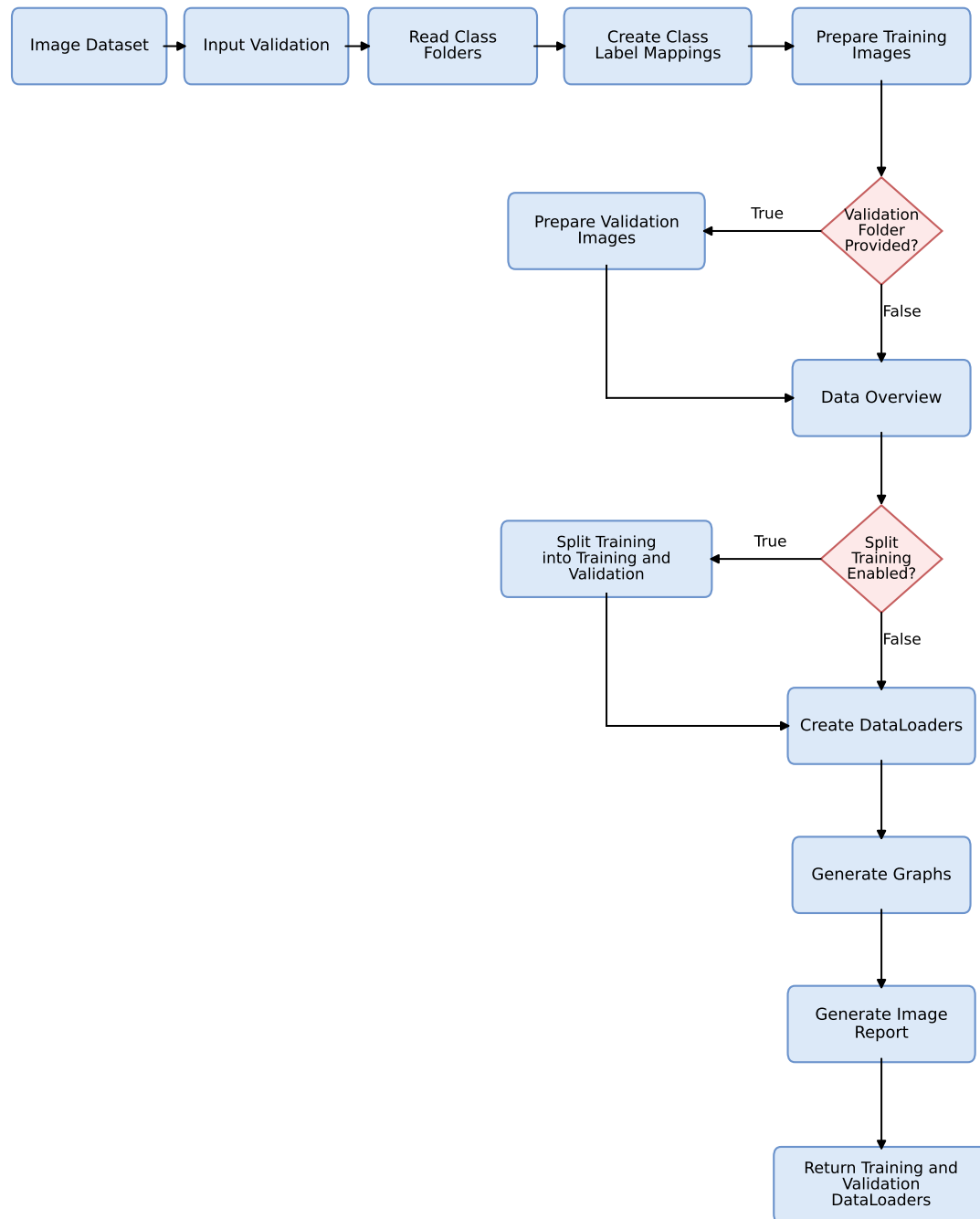


Figure 4.4: Image preprocessing workflow

The image classification preprocessing class expects a folder structure where each class has its own folder. It scans the folders, builds class to label and label to class mappings, validates images, optionally splits the training folder into training and validation

sets, applies resizing and augmentation, and returns PyTorch data loaders.

The validation folder is optional. If it is provided, its class folders must match the training classes. If no validation folder is provided, the module can split the training folder using the configured validation size. Invalid or unreadable image files are not used for training, but they are recorded in the report.

The image path saves:

- class to label mapping
- label to class mapping
- image size information
- report data about valid and invalid images.

The image report includes label summary graphs, random sample images from the folders, and sample images after passing through the data loader. These graphs are important because they show whether resizing, and augmentation settings are working as expected.

The class mappings are created from the sorted training class names. This means the same class name always receives the same numeric label as long as the class folder names are the same. The mappings are saved in `class2label_mapping.pkl` and `label2class_mapping.pkl`. The selected image size is also saved in `data_info.pkl`. For each class folder, the module checks every image path using the image validation utility. Valid images are stored with their numeric labels. Invalid images are skipped, but their paths and labels are saved in the report data so the user can know which files were removed. If no valid image exists, the module raises an error instead of returning an empty loader.

The `DataLoader` construction is also different between training and validation. The training loader uses shuffling and can apply augmentation. The validation loader does not use augmentation and does not shuffle, because validation should measure model behavior on stable data. The available augmentation settings include horizontal flip, vertical flip, rotation, brightness change, and contrast change, controlled by the probability and parameter values passed to the constructor.

Algorithm 4.3: Image training preprocessing flow from the implementation

Input: training image folder, optional validation folder

Output: training and validation `DataLoaders`

- 1: Read class folders from the training directory
- 2: Create class to label and label to class mappings
- 3: Save mappings and image size information
- 4: Validate image files and skip invalid images
- 5: Build the data overview for the report

- 6: Split training images if no validation folder is provided
 - 7: Create training dataset with resize and optional augmentation
 - 8: Create validation dataset without augmentation
 - 9: Build DataLoaders
 - 10: Generate label, sample, and loader graphs
 - 11: Generate the image preprocessing report
-

4.1.7 Image Inference Flow

The image testing class is used for new images after the training preprocessing artifacts already exist. It loads the saved image size and class mapping, validates the provided image paths, and creates a test dataset and DataLoader. If class names are provided, each class name is converted to its saved numeric label. If class names are not provided, the loader can still be used for prediction only mode.

Testing does not apply augmentation. It only applies the same resizing and tensor conversion rules needed to make images compatible with the trained model. Invalid images are returned separately, which is useful because the user can see which inputs were ignored instead of silently losing files.

4.1.8 Preprocessing Reports

The Auto Preprocessing report belongs to the Auto Preprocessing module because it explains what happened to the dataset during preprocessing. It is separate from the Accelera Pipe report, which explains pipeline branches and metric results.

For tabular data, `TabularPreprocessingReport` receives the `report_data` dictionary collected during preprocessing. The report contains:

- original and lowercased data previews
- dataframe structure, numerical statistics, and categorical statistics
- missing values, duplicate rows, and dataset shape
- target null handling and duplicate removal results
- train validation split sizes
- dropped columns and the reason for each dropped column
- EDA graphs generated for each feature column
- preprocessing decision for every feature
- processed samples of training and validation data
- feature importance scores and selected features.

For text data, the same report class is reused in text mode. It shows the text overview, text graph, target graph, cleaning and tokenization preview, TF-IDF step, and encoded target samples. This helps show how raw sentences became a sparse numerical matrix.

For image data, `ImagePreprocessingReport` is used instead. It shows class names, class-to-label mapping, number of valid images, invalid image counts, invalid image paths, split information, label summary graphs, random sample images, and samples after `DataLoader` preprocessing.

This report is useful because preprocessing is usually a hidden part of a machine learning project. With the report, the user can see the important changes instead of only receiving transformed arrays.

4.2 Accelera Pipe Module

4.2.1 Metric Node Wrappers

Metric nodes are part of the Accelera Pipe execution. They are responsible for calculating evaluation results after a model or clustering step produces predictions. The reporting module may display these results later, but the metric node itself belongs to the pipeline because it performs the metric calculation during the pipeline flow.

The concrete metric node wrappers are found in `wrappers/supervised_metric.py` and `wrappers/unsupervised_metric.py`. They are separated because supervised and unsupervised metrics need different inputs:

- `SupervisedMetric` is used when the true target values are available. The user provides `y_true`, then the node receives `y_pred` during execution. The wrapper checks that both arrays have the same number of rows, calls the selected metric function as `metric(y_true, y_pred, ...)`, and returns a metric result dictionary.
- `UnSupervisedMetric` is used when the metric depends on the input data and predicted labels, such as clustering metrics. The user sets `X`, then the node receives `y_pred`. The wrapper checks that the predicted labels match the number of rows in `X`, calls the selected metric function as `metric(X, y_pred, ...)`, and returns the same result dictionary format.

Both wrappers return a dictionary containing `metricname`, `result`, `plot_func`, `labels_name`, and `headers_name`. This shared output format makes the metric node easy to connect with later pipeline steps. If the user generates a pipeline report, the same dictionary can also be passed to the report display layer.

4.2.2 Pipeline Reporting

This part focuses only on the report classes related to `accelera_pipe`. These reports are different from the Auto Preprocessing report. The Auto Preprocessing report

explains data cleaning and dataset transformation, while the pipeline report explains the executed pipeline graph, branches, selected path, and metric results.

The pipeline reporting part has three main layers. `ReportBase` provides the shared HTML page structure. `GraphPipelineReport` groups and displays pipeline metric results. `GraphReport` adds the graph image, branch tables, and best branch summary.

4.2.2.1 Accelerera Report Base

`ReportBase` defines shared HTML report behavior. It stores the start and end HTML content, common styling, graph display helpers, and the `create_html_file` method that writes `report.html`. This class is important because different report classes can reuse the same structure instead of each one writing a full HTML page from the beginning. The base report also has a helper for showing saved graph images. It receives a folder path and image names, checks whether the graph files exist, and adds the images to the HTML content. This keeps graph display consistent for any report that needs to include saved image files.

4.2.2.2 Pipeline Metric Report

`GraphPipelineReport` is a pipeline reporting class, not the same as the base Accelerera report. It inherits from `ReportBase`, but its job is to format metric results produced by the pipeline execution. It receives a list of metric result dictionaries. Each result contains the metric name, the metric value, optional plotting function, labels, and headers.

The class groups results by metric name using `handle_metric`. After that, `metric_display` chooses the suitable display wrapper based on the result type:

- scalar numbers are displayed with `DisplaySingleNumber`. This wrapper is used for simple metric scores, such as accuracy or loss. It shows the metric ids and their values in a table, and also draws a small line graph so the user can compare the values visually.
- one dimensional arrays are displayed with `DisplayArraySingle`. This wrapper is used when the metric returns a list of values, such as per-class scores. It creates a table where each value has a label, and each row is connected to the metric id.
- multi dimensional arrays are displayed with `DisplayMultiArray` when there is no plotting function. This is useful for results such as matrices, where rows and columns need labels. If a plotting function is provided, `DisplayFigure` is used instead to save the plotted result as an image and show it in the report.
- dictionaries are displayed with `DisplayDict`. This wrapper converts dictionary results into HTML tables, so named values or grouped metric outputs can be read clearly.
- strings are displayed with `DisplayString`. This wrapper is used for text-based metric outputs, such as a classification report, and shows the result as formatted text.

- tuples are displayed with `DisplayTupleNotCurve` when there is no plotting function. This wrapper turns tuple elements into a table using labels and headers. If a plotting function is provided, `DisplayFigure` is used to display the tuple result as a figure.

4.2.2.3 Pipeline Graph Report

`GraphReport` is another pipeline report class. It extends `GraphPipelineReport`, but it adds graph-structure reporting. It reads the serialized XML graph file, extracts nodes and edges, renders a Graphviz image, finds branches through the graph, displays all branches, and also shows the best branch based on nodes marked as selected in the final path.

The generated graph image labels each node using node id, node type, and node name. Selected nodes are drawn in red, while the other nodes are drawn in blue. Metric nodes are also collected so that the metric summary can be matched with the graph branch where the metric was produced.

Algorithm 4.4: Pipeline graph report generation flow

Input: XML graph path, metric results

Output: HTML pipeline report and graph image

- 1: Parse the XML graph file
 - 2: Read nodes from XML layers
 - 3: Mark selected nodes and collect metric node ids
 - 4: Read edges and build graph connections
 - 5: Render the pipeline graph image with Graphviz
 - 6: Extract all branches until metric nodes are reached
 - 7: Group branches that share the same path
 - 8: Display all branches and the selected best branch
 - 9: Group metric results and format them by result type
 - 10: Write report.html
-

4.3 Benchmark Platform Module

4.3.1 Module Purpose

The purpose of the Benchmarking module is to support learning and improvement inside Accelera. It gives users a place where they can try a machine learning task, submit their predictions, see their score, and compare their result with other submissions. By viewing the leaderboard and the submitted repository links, users can learn from the techniques used by others, understand which approaches perform better, and improve their own solution before submitting again.

The module also keeps this learning process fair because all users work on the same

benchmark task and are evaluated using the same hidden ground truth file and metric configuration. A benchmark creator defines the problem, dataset links, test data, target column, and evaluation metric. Other users submit their prediction files, and the system validates the files, calculates the score automatically, and displays the results on the leaderboard.

The module is implemented in the `benchmark` folder. It has a Node.js and Express backend, MongoDB schemas using Mongoose, Python scripts for file validation and scoring, and a React frontend for benchmark creation, submission, and leaderboard screens.

4.3.2 Functional Description

When creating a benchmark, the user provides a title, description, problem type, target column, dataset link, test dataset link, ground truth target link, and evaluation metric. The module checks that the test dataset does not include the target column and that the target file includes the required target column.

When submitting to a benchmark, the user provides a prediction file and a repository link. The scoring process loads the target file and prediction file, checks that they have the same number of rows, and computes the selected metric. The submission is then saved and displayed in the leaderboard.

4.3.3 Backend Architecture

The backend server is started from `benchmark/backend/server.js`. It loads configuration, connects to MongoDB, enables CORS and JSON parsing, and mounts the main routes:

- `/metrics` for metric creation and listing
- `/benchmark` for benchmark creation and browsing
- `/submission` for leaderboard submissions.

The backend uses schemas for benchmarks, metrics, and submissions. The route files handle benchmark browsing and creation, metric browsing and creation, and submission scoring. The Python scripts are called from the backend when the system needs to validate uploaded links or calculate a submission score.

4.3.4 Database Models

The main benchmark data is stored in `benchmark`, `metric`, and `submission` collections. The `metric` collection stores the display name, the scikit learn metric function name, needed parameters, problem type, and whether higher or lower values are better.

The `benchmark` collection stores the benchmark title, description, target column, dataset link, test set link, target label link, problem type, evaluation metric, metric parameters,

creation date, and creator. The submission collection stores benchmark id, submitter id, repository link, prediction file link, submission date, and score. It also has an index on benchmark id and score to support leaderboard access.

4.3.5 Main Backend Components

Table 4.4: Benchmarking module components

Component	Responsibility
Benchmark management	Create, list, filter, view, and delete benchmark tasks.
Metric management	Define metrics, problem type, parameters, and whether higher or lower score is better.
Submission management	Store prediction submissions, repository links, scores, submitter, and submission date.
Validation scripts	Check benchmark files and submission files before scoring.
Scoring scripts	Compute the selected scikit learn metric on predictions and ground truth.
Leaderboard interface	Display submissions ordered by the metric direction.

4.3.6 Benchmark Creation Flow

The benchmark creation endpoint is implemented in `routes/benchmark.js`. The backend first normalizes the problem type and checks that it is either classification or regression. It also checks that the benchmark title is unique. Dataset and file links are validated, and Google Drive file links are required for the test file and target file. After link validation, the backend calls the Python validation script through `run_python`. The script downloads or retrieves the test and target files, then checks two important rules:

1. the test dataset must not contain the target column
2. the target file must contain the target column.

The current validation script also checks that the target column is numeric. If the validation succeeds, the benchmark is saved in MongoDB.

Algorithm 4.5: Benchmark creation flow

Input: benchmark form data

Output: saved benchmark or validation error

- 1: Receive benchmark creation request
- 2: Validate problem type

- 3: Check title uniqueness
 - 4: Validate dataset, test set, and target label links
 - 5: Run `validate_user_links.py`
 - 6: If the test file contains the target column, reject benchmark
 - 7: If the target file does not contain the target column, reject benchmark
 - 8: Save benchmark document in MongoDB
-

4.3.7 Metric Management

Metrics are managed through `routes/metrics.js`. A metric is created by providing a display name, scikit learn metric function name, problem type, needed parameters, and `whichBetter`. The `whichBetter` value must be either `higher` or `lower`. This value is important because it controls leaderboard sorting.

The system prevents duplicate metrics with the same name, problem type, and scikit learn metric name. It also prevents deleting a metric if at least one benchmark still uses it.

4.3.8 Submission and Scoring Flow

The submission endpoint is implemented in `routes/submissions.js`. The module prevents the same submitter from creating more than one submission for the same benchmark. This is done by checking for an existing submission with the same benchmark id and submitter id. If the submitter wants to change the prediction file later, the update endpoint is used instead.

For each submission, the backend validates the prediction file link. Then it calls `get_score.py`. The Python script retrieves the true labels and submitted predictions, checks that the files have the same number of rows, checks that the prediction file contains the target column, and then calls the selected metric from `sklearn.metrics`.

Algorithm 4.6: Submission scoring flow

Input: benchmark target file, submitted prediction file, metric name

Output: score or validation error

- 1: Retrieve true target file
 - 2: Retrieve submitted prediction file
 - 3: Check that both files have the same number of rows
 - 4: Check that submitted file contains the target column
 - 5: Load metric function from `sklearn.metrics`
 - 6: Compute `metric(true_target, predicted_target, metric_parameters)`
 - 7: Return score to the backend as JSON
 - 8: Save submission with score
-

The scoring algorithm is kept in Python because the project already uses Scikit learn metrics there. The backend sends the needed file links, target column, metric name, and metric parameters. The Python script returns either a score or an error message.

4.3.9 Leaderboard Ordering

Leaderboard ordering is handled by the submission route. When submissions are requested for a benchmark, the backend reads the benchmark metric and checks `whichBetter`. If the metric is better when higher, submissions are sorted by descending score. If the metric is better when lower, submissions are sorted by ascending score. This lets the same module support metrics such as accuracy and F1 score, where higher is better, and metrics such as mean absolute error, where lower is better.

4.3.10 Frontend Interface

The frontend is implemented with React under `benchmark/frontend/src`. The relevant pages for benchmarking are:

- `home.jsx`: entry page with links to benchmarks and metrics
- `benchmarks.jsx`: lists benchmarks and filters them by user or problem type
- `createBenchmark.jsx`: form for benchmark creation
- `metrics.jsx` and `createMetric.jsx`: metric listing and metric creation
- `displayBenchmark.jsx`: shows benchmark details and
- `leaderBoard.jsx`: displays submissions and allows adding, updating, or deleting submissions when permitted.

4.3.11 Benchmark Constraints

The module has several important constraints:

- it currently supports classification and regression tasks
- datasets and prediction files are provided as links
- metrics must map to valid scikit learn metric functions

Chapter 5: System Testing and Verification

5.1 Testing Setup

The modules were tested using real datasets, saved preprocessing artifacts, generated reports, and benchmark workflow checks. The Auto Preprocessing results were evaluated by training machine learning models on the transformed outputs. Reporting was checked by generating HTML and graph outputs. Benchmarking was checked through benchmark creation, metric handling, submission validation, scoring, and leaderboard behavior.

5.2 Testing Plan and Strategy

The testing strategy focuses on checking each module separately and making sure that its output can be used correctly by the next step in the workflow. The tests were designed around the main responsibility of each part. For Auto Preprocessing, the important check was that the module can process the training data, save the fitted preprocessing artifacts, and reuse them later on test data. For the reporting part, the main check was that the pipeline information, branches, graphs, and metric results are collected correctly and displayed in a readable report. For the Benchmarking module, the main check was that benchmark files and user submissions are validated correctly, then scored using the same target file and metric configuration for all users.

5.3 Module Testing

5.3.1 Benchmark Platform Module

The Benchmark Platform module was evaluated using manual functional testing to check its main features from the user side. The testing focused on creating benchmark tasks, validating dataset links, submitting prediction files, calculating scores, and displaying the results in the leaderboard.

The leaderboard was tested by submitting different results and checking whether they were ordered correctly based on the selected evaluation metric. This included metrics where higher values are better and metrics where lower values are better.

The scoring process was also tested manually by using different target and prediction files. The platform was checked to make sure it rejects invalid submissions, such as files with different row counts or missing target columns. These tests showed that the benchmark platform can handle result submission, calculate scores, validate files, and

display leaderboard results correctly.

5.3.2 Auto Preprocessing Module

Most of the preprocessing functions were tested using unit tests. These tests were used to check validation rules, artifact saving, and feature transformation. Real datasets were also used to test the system in a more practical way. These datasets were collected from Kaggle for tabular, text, and image data.

5.3.2.1 How the Auto Preprocessing testing works

The Auto Preprocessing testing was done as a practical full-flow test. The idea was to run preprocessing on real datasets, then pass the processed output to a model. This means the test does not only check that preprocessing runs without errors, but also checks that the returned data can actually be used for training, validation, and prediction.

For tabular datasets, the test starts by loading each dataset with its target column and problem type. The preprocessing flow splits the data, fits the needed encoders and scalers, transforms the features and target, applies feature selection, and saves the preprocessing artifacts. After that, a machine learning model is trained on the processed training data and evaluated on the processed validation data. Classification datasets are evaluated using Macro F1-score, and regression datasets are evaluated using the R-squared score. The same datasets are also processed by a baseline automated cleaning tool, so the final scores can be compared and the Accelera output can be checked against an existing preprocessing approach.

For text datasets, the test checks that raw text can be converted into features that a classifier can understand. The preprocessing flow receives the text column and target column, cleans the text, tokenizes it, applies TF-IDF, encodes the labels, and returns processed training and validation data. A classifier is then trained on the transformed text features and evaluated using Macro F1-score. This confirms that the text preprocessing output is numeric, consistent, and usable for a normal machine learning model.

For image classification, the test checks the full image preparation flow. The preprocessing reads images from class folders, validates image files, skips corrupted images, creates class-to-label mappings, resizes the images, applies augmentation when enabled, and returns PyTorch DataLoaders. A neural network model is then trained using these loaders. The inference path is also checked by loading the saved model and applying the testing preprocessing to new images. This verifies that the saved image size and class mapping can be reused during testing, not only during training.

The main concept in all three tests is to check preprocessing through the next step in the machine learning pipeline. If the tabular arrays, TF-IDF matrices, labels, image tensors, or saved artifacts were wrong, the model training or evaluation step would fail or give invalid results.

Table 5.1: Datasets Used in Module Testing

Dataset Name	Shape	Dataset Type	Problem Type
startup_founder_burnout_2026	(50000, 29)	Tabular	Classification
healthy_diet_calorie_intake	(6000, 15)	Tabular	Classification
gaming_addiction	(250, 49)	Tabular	Classification
student_exam_performance_dataset	(10000, 17)	Tabular	Classification
heart	(1025, 14)	Tabular	Classification
Dataset_spine	(310, 7)	Tabular	Classification
diabetes_dataset	(100000, 31)	Tabular	Classification
student_placement_synthetic	(100000, 18)	Tabular	Classification
Titanic-Dataset	(891, 12)	Tabular	Classification
customer_purchase_data	(1500, 9)	Tabular	Classification
global_ai_jobs	(90000, 35)	Tabular	Regression
CarPrice_Assignment	(205, 26)	Tabular	Regression
Housing	(545, 13)	Tabular	Regression
PetImages	(24998,)	Image	Classification
DailyDialog	(11327, 2)	Text	Classification
TestReviews	(4321, 2)	Text	Classification

Table 5.2: Models and Evaluation Metrics Used in Module Testing

Dataset Name	Model Used	Evaluation Metric	Accelera Result
startup_founder_burnout_2026	RandomForest	Macro F1-score	0.9999
healthy_diet_calorie_intake	RandomForest	Macro F1-score	1.0000
gaming_addiction	RandomForest	Macro F1-score	1.0000
student_exam_performance_dataset	RandomForest	Macro F1-score	0.8573
heart	RandomForest	Macro F1-score	0.7860
Dataset_spine	RandomForest	Macro F1-score	0.7840
diabetes_dataset	RandomForest	Macro F1-score	0.9985
student_placement_synthetic	RandomForest	Macro F1-score	0.9986
Titanic-Dataset	RandomForest	Macro F1-score	0.7906
customer_purchase_data	RandomForest	Macro F1-score	0.9385
global_ai_jobs	RandomForest	R-squared Score	0.9227
CarPrice_Assignment	RandomForest	R-squared Score	0.9562
Housing	RandomForest	R-squared Score	0.6037
PetImages	ResNet model	Accuracy	0.9700
DailyDialog	XGBClassifier	Macro F1-score	0.9000
TestReviews	XGBClassifier	Macro F1-score	0.6080

These results show that the preprocessing output can be used by different models after transformation. The tabular results were strong in many datasets, especially the larger classification datasets and the car price regression dataset. The lower scores on datasets such as TestReviews show that preprocessing alone does not solve every modeling problem. Some datasets still need better features, model tuning, or more suitable models.

5.4 Comparative Results to Previous Work

5.4.1 Auto Preprocessing

Auto Preprocessing was compared with AutoCleanML to check the performance of the proposed preprocessing approach. The comparison was done using several tabular datasets collected from Kaggle. The same datasets and the same machine learning models were used for both tools. The comparison was applied only on tabular datasets because AutoCleanML mainly works with tabular data. For classification datasets, the Macro F1-score was used as the evaluation metric. For regression datasets, the R-squared score was used to measure how well the model predicts the target values. After preprocessing the datasets, the output data was tested using four machine learning models. For classification problems, Logistic Regression, Random Forest, Decision

Tree, and KNN were used. For regression problems, Linear Regression, Random Forest, Decision Tree, and KNN were used. This helped test the preprocessing output with different model types, not only one model. The results were recorded and compared between Accelera and AutoCleanML for each selected dataset and model. The purpose of this comparison was to check whether Accelera can produce preprocessing results that are close to, or better than, an existing preprocessing tool.

Table 5.3: Classification comparison using Logistic Regression and Random Forest

Dataset	Logistic Regression		Random Forest	
	AutoCleanML	Accelera	AutoCleanML	Accelera
Startup founder burnout	1.0000	1.0000	0.9999	0.9999
Healthy diet calorie intake	0.9917	0.9890	1.0000	1.0000
Gaming addiction	0.8937	0.8937	0.9646	1.0000
Student exam performance	0.8667	0.8657	0.8579	0.8573
Heart	0.8019	0.8007	0.7689	0.7860
Spine dataset	0.8691	0.7943	0.8360	0.7840
Diabetes dataset	0.9995	0.9985	0.9996	0.9985
Student placement synthetic	0.5657	0.5879	0.6241	0.9986
Titanic dataset	0.7734	0.7864	0.7906	0.7906
Customer purchase data	0.8484	0.8375	0.9386	0.9385

Table 5.4: Classification comparison using Decision Tree and KNN

Dataset	Decision Tree		KNN	
	AutoCleanML	Accelera	AutoCleanML	Accelera
Startup founder burnout	1.0000	1.0000	0.9824	0.9881
Healthy diet calorie intake	1.0000	1.0000	0.9379	0.9184
Gaming addiction	1.0000	1.0000	0.6905	0.8512
Student exam performance	0.7987	0.8063	0.8423	0.8338
Heart	0.7213	0.7704	0.7689	0.7860
Spine dataset	0.7686	0.7632	0.8239	0.8042
Diabetes dataset	0.9995	0.9972	0.9925	0.9980
Student placement synthetic	0.5827	0.9968	0.5995	0.8348
Titanic dataset	0.6979	0.7877	0.7203	0.7562
Customer purchase data	0.8486	0.8449	0.8544	0.8806

Table 5.5: Regression comparison using Linear Regression and Random Forest

Dataset	Linear Regression		Random Forest	
	AutoCleanML	Accelera	AutoCleanML	Accelera
Global AI jobs	0.9022	0.7498	0.9288	0.9227
Car price assignment	0.8886	0.8943	0.9535	0.9562
Housing	0.6506	0.6482	0.6087	0.6037

Table 5.6: Regression comparison using Decision Tree and KNN

Dataset	Decision Tree		KNN	
	AutoCleanML	Accelera	AutoCleanML	Accelera
Global AI jobs	0.8640	0.8412	0.7586	0.7510
Car price assignment	0.9269	0.8825	0.7905	0.7478
Housing	0.3819	0.3634	0.5312	0.5336

From the classification and regression results, Accelera preprocessing achieved competitive performance compared to AutoCleanML. The comparison was tested using different types of models, including linear models, tree based models, Random Forest, and KNN. In many cases, the results of Accelera were very close to AutoCleanML, and in some cases Accelera achieved better scores.

These results show that the implemented preprocessing steps in Accelera are able to prepare data in a valid way for different machine learning techniques. The performance was not limited to one specific model type, but appeared across several models and datasets. This means that Accelera successfully reached results close to an existing automated preprocessing tool, while still using its own preprocessing approach.

However, the results also show that no preprocessing tool gives the best score in all cases. The final performance can still depend on the dataset, the problem type, and the model used after preprocessing.

5.4.2 Pipeline Reporting Module Testing

The graph report is exercised through examples that build branch-based workflows, serialize the graph to XML, and run `GraphReport`. The expected result is a `report.html` file and a graph image showing the serialized graph structure. The report also checks that metrics are displayed according to their result type.

Reference

- [1] Matthieu Komorowski, Dominic C. Marshall, Justin D. Saliccioli, and Yves Crutain. “Exploratory data analysis.” In *Secondary Analysis of Electronic Health Records*, pages 185–203. Springer, 2016.
- [2] Mohammad Hossin and Md Nasir Sulaiman. “A review on evaluation metrics for data classification evaluations.” *International Journal of Data Mining and Knowledge Management Process*, 5(2):1, 2015.
- [3] Navin Sabharwal and Amit Agrawal. “Introduction to Natural Language Processing.” In *Hands-on Question Answering Systems with BERT: Applications in Neural Networks and Natural Language Processing*, pages 1–14. Springer, 2021.

Chapter 6: Conclusions and Future Work

6.1 Faced Challenges

In Auto Preprocessing, one challenge was that there are many possible preprocessing techniques, such as different imputation methods, scaling methods, encoding methods, and feature selection methods. It was not practical to try every technique for every dataset, so the module needed a rule-based approach that chooses techniques expected to work well for many datasets. The challenge was to make these rules general enough to be useful, but still simple and stable enough to run automatically.

In the reporting part, another challenge was handling the different outputs that can come from pipeline execution. Scikit-learn metrics may return single values, per-class values, arrays, matrices, or text reports. The reporting code needed to display these different formats clearly in one report without losing important information. It was also difficult to organize graph information, branches, selected paths, and metric summaries together in a readable way.

6.2 Gained Experience

This work provided practical experience in building machine learning support modules that are used before, during, and after model training. In Auto Preprocessing, I gained experience in choosing suitable preprocessing rules for different types of data, handling missing values, encoding categorical features, scaling numerical features, preparing text and image inputs, and saving fitted artifacts so the same transformations can be reused during testing.

In the pipeline reporting part, I gained experience in collecting execution information from a machine learning workflow and presenting it in a clear form. This included working with graph structure, pipeline branches, selected paths, metric nodes, and different metric output formats. It also improved my understanding of how reports can help users inspect the result of a pipeline instead of depending only on terminal output. In the Benchmarking module, I gained experience in backend workflow design, database schemas, file validation, scoring submissions, and leaderboard presentation. I also learned how important it is to keep evaluation fair by using the same hidden target file and metric configuration for all submissions. Testing these parts with real datasets helped me understand how implementation details affect the full machine learning workflow.

6.3 Conclusions

This report documented my individual work in the Auto Preprocessing module, the Accelera Pipe module, and the Benchmarking module in Accelera. The Auto Preprocessing module prepares tabular, text, and image datasets before model training. It handles repeated steps such as cleaning, splitting, encoding, scaling, text vectorization, image loading, and artifact saving. The most important point in this module is consistency, because the fitted preprocessing objects saved during training are reused later during testing.

The Accelera Pipe module includes metric nodes and pipeline reporting. The metric nodes calculate supervised or unsupervised evaluation results during the pipeline flow. The reporting part helps make pipeline outputs easier to understand. Instead of leaving the user with only terminal output, it generates HTML reports, graph images, tables, and metric displays. This is useful when the workflow contains branches or different metric result types. The Benchmarking module adds a fair comparison workflow by using the same dataset, target labels, and metric configuration for submitted predictions.

From the testing results, the preprocessing outputs were usable by machine learning and deep learning models for tabular, text, and image tasks. The generated reports also helped explain what happened during preprocessing and pipeline execution. Together, these modules make the data science workflow more organized, more reusable, and easier to inspect. The implementation still has limitations, but it provides a clear base for future improvements.

6.4 Future Work

6.4.1 Design Advantages

The main advantage of Auto Preprocessing is consistency. The same fitted objects used during training are saved and loaded during testing. This reduces the chance of data leakage and mismatched transformations. Another advantage is that the module supports more than one data type, so the same project can prepare tabular, text, and image datasets using suitable preprocessing steps for each type. The preprocessing reports also help with data understanding and exploratory data analysis by showing summaries, missing values, feature distributions, correlations, and the preprocessing decisions applied to the dataset.

The main advantage of the Reporting module is explainability. The user can see the executed pipeline graph, branches, selected paths, and metric results in one place. This makes the pipeline output easier to understand, especially when the workflow contains more than one model path or different metric result formats.

The Benchmarking module helps make comparison fairer because all submissions are evaluated using the same ground truth and metric configuration.

6.4.2 Current Limitations

There are also limitations:

- the preprocessing rules are fixed heuristics and may not be optimal for every dataset
- the tabular preprocessing path assumes ordinary structured data and may need extensions for time series or special domain data
- text preprocessing uses classical TF-IDF, not transformer based language models
- benchmark files are provided through links, so file availability depends on external storage and

6.4.3 Possible Improvements

Future improvements can include adaptive preprocessing based on the model type, better handling of imbalanced datasets, more advanced text preprocessing, and server-side benchmark file storage.

Chapter A: User Guide

This chapter gives simple examples for calling the main classes from the code. The examples are small, but they show the normal constructor calls and the main method used to start preprocessing or report generation.

A.1 Using Tabular Auto Preprocessing

For tabular data, the user loads the dataset as a Pandas DataFrame, chooses the target column, chooses the problem type, and gives an output folder. The output folder stores the fitted preprocessing objects and the report.

Code Example A.1: Calling tabular training preprocessing

```
import pandas as pd
from accelera.src.auto_preprocessing.core.classical_training_preprocessing import (
    ClassicalTrainingPreprocessing,
)

df = pd.read_csv("dataset.csv")

preprocessor = ClassicalTrainingPreprocessing(
    df=df,
    target_col="target",
    problem_type="classification",
    folder_path="outputs/tabular_run",
    val_size=0.2,
    random_state=42,
    is_report=True,
)

X_train, y_train, X_val, y_val = preprocessor.common_preprocessing()
```

For regression, the main change is the problem type and target column.

Code Example A.2: Calling tabular regression preprocessing

```
preprocessor = ClassicalTrainingPreprocessing(
    df=df,
    target_col="price",
    problem_type="regression",
    folder_path="outputs/regression_run",
)
```

For test data, the user should pass the same folder path. This makes the test data use the saved training transformers instead of fitting new transformers.

Code Example A.3: Calling tabular testing preprocessing

```
import pandas as pd
from accelera.src.auto_preprocessing.core.classical_testing_preprocessing import (
    ClassicalTestingPreprocessing,
)

test_df = pd.read_csv("test.csv")

test_preprocessor = ClassicalTestingPreprocessing(
    df=test_df,
    folder_path="outputs/tabular_run",
)

X_test, y_test = test_preprocessor.common_preprocessing()
```

A.2 Using Text Auto Preprocessing

For text data, the user provides the text column and target column. The module cleans the text, applies TF-IDF, encodes the labels, and saves the needed objects.

Code Example A.4: Calling text training preprocessing

```
import pandas as pd
from accelera.src.auto_preprocessing.core.text_training_preprocessing import (
    TextTrainingPreprocessing,
)

df = pd.read_csv("reviews.csv")

text_preprocessor = TextTrainingPreprocessing(
    df=df,
    target_col="label",
    text_col="review",
    folder_path="outputs/text_run",
    tfidf_max_features=500,
    is_report=True,
)

X_train, y_train, X_val, y_val = text_preprocessor.common_preprocessing()
```

For testing text data, the user again passes the same folder path. If the test data does not contain the target column, the class returns the transformed text features and `None` for labels.

Code Example A.5: Calling text testing preprocessing

```
from accelera.src.auto_preprocessing.core.text_testing_preprocessing import (
    TextTestingPreprocessing,
)

test_text = pd.read_csv("reviews_test.csv")
```

```
text_test_preprocessor = TextTestingPreprocessing(
    df=test_text,
    folder_path="outputs/text_run",
)

X_test, y_test = text_test_preprocessor.common_preprocessing()
```

A.3 Using Image Auto Preprocessing

For image preprocessing, the dataset should be arranged as class folders. For example:

```
Training/
  Cats/
  Dogs/
```

The user passes the training folder and output folder. If there is no separate validation folder, the module can split the training folder.

Code Example A.6: Calling image training preprocessing

```
from accelera.src.auto_preprocessing.core.classification_image_training_preprocessing
    import (
        ClassificationImageTrainingPreprocessing,
    )

image_preprocessor = ClassificationImageTrainingPreprocessing(
    training_folder_images="data/PetImages/Training",
    folder_path="outputs/image_run",
    split_training=True,
    val_size=0.2,
    images_size=(224, 224),
    batch_size=16,
    augment=True,
)

train_loader, val_loader = image_preprocessor.common_preprocessing()
```

For image testing, the user provides a list of image paths. Class names are optional, but if they are provided, they must match the saved training classes.

Code Example A.7: Calling image testing preprocessing

```
from accelera.src.auto_preprocessing.core.classification_image_testing_preprocessing
    import (
        ClassificationImageTestingPreprocessing,
    )

image_paths = ["test/cat1.jpg", "test/dog1.jpg"]
class_names = ["Cats", "Dogs"]
```

```

image_test_preprocessor = ClassificationImageTestingPreprocessing(
    image_paths=image_paths,
    image_class_names=class_names,
    folder_path="outputs/image_run",
    batch_size=4,
)

test_loader, invalid_images = image_test_preprocessor.common_preprocessing()

```

A.4 Reading Auto Preprocessing Reports

After running preprocessing with `is_report=True`, the user opens the generated `report.html` file from the output folder. The report shows the dataset overview, missing values, duplicate rows, dropped columns, graphs, preprocessing decisions, selected features, and processed data samples.

A.5 Using Pipeline Reports

To generate a pipeline report, the pipeline graph should be serialized first. Then the user passes the XML graph path, output folder, and metric results to `GraphReport`.

Code Example A.8: Calling the pipeline graph report

```

from accelera.src.accelera_pipe.wrappers.graph_report import GraphReport

report = GraphReport(
    folderpath="outputs/pipeline_report",
    xmlpath="outputs/pipeline_graph.xml",
    results=metric_results,
)

report.execute()

```

The output is a `report.html` file. It contains the graph image, branch tables, selected branch, and metric summary.

A.6 Using the Benchmark Platform

The benchmark platform is used from the web interface. The benchmark creator enters the title, problem type, target column, dataset link, test-set link, target-label link, and evaluation metric. A user submits a prediction file link and repository link. The backend calls the Python scoring script, saves the score, and displays the result in the leaderboard. The prediction file must contain the target column expected by the benchmark. Also, the number of rows in the prediction file must match the number of rows in the target label file.